

Adaptive Business Intelligence

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Springer

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To Adam, Ewa, and Arthur.
Z. M.

*To Ana, my family and my friends:
May Love, Knowledge and Good Luck guide your way!*
M. S.

To my parents, children, and loving wife Luiza.
M. M.

To my parents and my wife Larisa.
C. C.

Preface

“My name is Sherlock Holmes. It is my business to know what other people do not know.”

The Adventure of the Blue Carbuncle

“What do you think of it, Watson?”

“A masterpiece. You have never risen to a greater height.”

The Adventure of the Bruce-Partington Plans

Since the computer age dawned on mankind, one of the most important areas in information technology has been that of “decision support.” Today, this area is more important than ever. Working in dynamic and ever-changing environments, modern-day managers are responsible for an assortment of far-reaching decisions: *Should the company increase or decrease its workforce? Enter new markets? Develop new products? Invest in research and development?* The list goes on. But despite the inherent complexity of these issues and the ever-increasing load of information that business managers must deal with, all these decisions boil down to two fundamental questions:

- What is likely to happen in the future?
- What is the best decision right now?

Whether we realize it or not, these two questions pervade our everyday lives – both on a personal and professional level. When driving to work, for instance, we have to make a traffic prediction before we can choose the quickest driving route. At work, we need to predict the demand for our product before we can decide how much to produce. And before investing in a foreign market, we need to predict future exchange rates and economic variables. It seems that regardless of the decision being made or its complexity, we first need to make a prediction of what is likely to happen in the future, and then make the best decision based on that prediction. This fundamental process underpins the basic premise of *Adaptive Business Intelligence*.

Simply put, Adaptive Business Intelligence is the discipline of combining prediction, optimization, and adaptability into a system capable of answering these two fundamental questions: *What is likely to happen in the future?* and *What is the best decision right now?* To build such a system, we first need to understand the methods and techniques that enable prediction, optimization, and adaptability. At first blush, this subject matter is nothing new, as hundreds of books have already been written on business intelligence, data mining and prediction methods, optimization techniques,

and so forth. However, none of these books has explained how to combine these various technologies into a software system that is capable of predicting, optimizing, and adapting. This text is the first on the subject.

When we set out to write *Adaptive Business Intelligence*, we had three important objectives in mind: First of all, we wanted to explain why the future of the business intelligence industry lies in systems that can make decisions, rather than tools that produce detailed reports. As most business managers now realize, there is a world of difference between having good knowledge and detailed reports, and making smart decisions. Michael Kahn, a technology reporter for Reuters in San Francisco, makes a valid point in the January 16, 2006 story entitled “Business intelligence software looks to future”:

“But analysts say applications that actually answer questions rather than just present mounds of data is the key driver of a market set to grow 10 per cent in 2006 or about twice the rate of the business software industry in general.

‘Increasingly you are seeing applications being developed that will result in some sort of action,’ said Brendan Barnacle, an analyst at Pacific Crest Equities. ‘It is a relatively small part now, but it is clearly where the future is. That is the next stage of business intelligence.’”

We could not agree more.

Second, we wanted to explain the principles behind many prediction methods and optimization techniques in simple terms, so that any business manager could grasp them. Even though most business managers have a limited technology background, they should not be intimidated by terms such as “artificial neural networks,” “fuzzy logic,” “evolutionary algorithms,” “ant systems,” or “agent-based modeling.” They should understand the strengths and weaknesses of these methods and techniques, their operating principles, and applicability. Armed with such knowledge, business managers will be in a better position to control the application of these methods and techniques in their respective organizations.

And, third, we wanted to underscore the enormous applicability of Adaptive Business Intelligence to many real-world business problems, ranging from demand forecasting and scheduling, to fraud detection and investment strategies. From a high-level perspective, most of these business problems have similar characteristics, and the application of Adaptive Business Intelligence can provide significant benefits and savings.

To facilitate the discussion in this book, we have divided the chapters into three parts that correspond to the three objectives listed above. In Part I, we present the fundamental ideas behind Adaptive Business Intelligence, and explain the different roles that prediction, optimization, and adaptability play in producing near-optimal decisions. We also discuss the characteristics that many business problems have in common, and why these characteristics increase the complexity of the problem-solving exercise. Furthermore, we introduce a particular distribution problem that is used throughout the text as a running example. Given that the prediction and optimization issues in this example are common to most business problems, it should be relatively easy for the reader to extrapolate this example to many other business domains.

Because countless texts have already been written on the subject of database technologies, data warehousing, online analytical processing, reporting, and the like, we saw little point in rehashing the tools and techniques that are routinely used to access, view, and manipulate organizational data. Instead, Part II of the book discusses the various prediction methods and optimization techniques that can be used to develop an Adaptive Business Intelligence system. The distribution example is continued throughout these chapters, effectively highlighting the strengths and weaknesses of each method and technique. Each chapter in Part II is concluded by a *Recommended Reading* section that provides suggestions for readers who want to learn more about particular methods or techniques.

Part III begins with a chapter on hybrid systems and adaptability, explaining how to “combine” the various methods and techniques discussed in Part II, and how the component of adaptability can be added to the final design. In the remaining chapters of the book, we discuss the definitive solution to the distribution problem that was used throughout the text, as well as the application of Adaptive Business Intelligence to several other complex business problems.

Without a doubt, we believe that business managers of all levels would benefit from this text. Anyone who makes operational and strategic decisions – whether on the factory floor or in the boardroom – will find this book invaluable for understanding the science and technology behind better predictions and decisions. We hope that readers will enjoy reading the book as much as we enjoyed writing it, and that they will profit from it.

We would like to thank everyone who made this book possible, and who took the time to share their thoughts and comments on the subject of Adaptive Business Intelligence. In particular, we would like to express our gratitude to SolveIT Software’s scientific advisory board, which includes Hussein Abbass, Valerio Aimale, Jürgen Branke, Mike Brooks, Carlos Coello, Ernesto Costa, Kalyanmoy Deb, Gusz Eiben, Fred Glover, Philip Hingston, Jong-Hwan Kim, Bob McKay, Marek Michalewicz, Masoud Mohammadian, Pablo Moscato, Michael Rumsewicz, Marc Schoenauer, Alice Smith, Russel Stonier, Lyndon While, Xin Yao, and Jacek Zurada. Our special appreciation also goes to two anonymous reviewers who provided us with many insights and useful suggestions, and to Ronan Nugent, who did a wonderful job of editing this book and helping us make the entire project a success.

Lastly, we would like to thank the most famous fictional detective of all time, Sherlock Holmes, for providing us with the entertaining quotes at the beginning of each chapter. Mr. Holmes remains one of the most famous problem-solvers of all time, and his methodology is based on prediction (“*It is a capital mistake to theorize before you have all the evidence*”), optimization (“*... I had best proceed on my own lines, and then clear the whole matter up once and for all*”), and adaptability (“*I have devised seven separate explanations ... But which of these is correct can only be determined by the fresh information, which we shall no doubt find waiting for us*”). Needless to say, his methodology bears a striking resemblance to Adaptive Business Intelligence! Enjoy.

Adelaide, Australia
October 2006

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Part I: Complex Business Problems

1 Introduction

“Like all Holmes’s reasoning, the thing seemed simplicity itself when it was once explained.”

The Stock-Broker’s Clerk

“Before turning to ... matter which present the greatest difficulties, let the inquirer begin by mastering more elementary problems.”

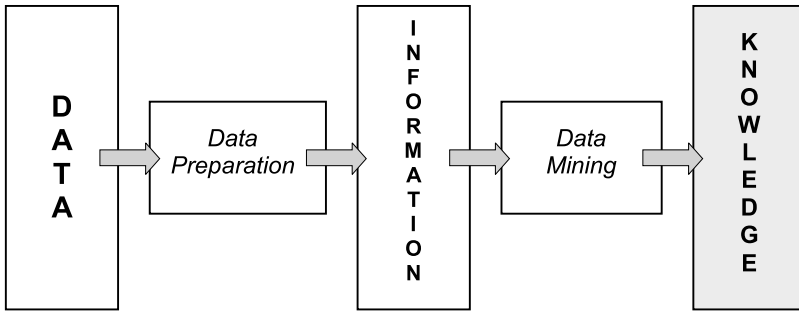
A Study in Scarlet

“The answer to my problem is hidden in my data... but I cannot dig it up!” This popular statement has been around for years as business managers gathered and stored massive amounts of data in the belief that they contain some valuable insight. But business managers eventually discovered that raw data are rarely of any benefit, and that their real value depends on an organization’s ability to analyze them. Hence, the need emerged for software systems capable of retrieving, summarizing, and interpreting data for end-users.

This need fueled the emergence of hundreds of *business intelligence* companies that specialized in providing software systems and services for extracting *knowledge* from raw data. These software systems would analyze a company’s operational data and provide knowledge in the form of tables, graphs, pies, charts, and other statistics. For example, a business intelligence report may state that 57% of customers are between the ages of 40 and 50, or that product X sells much better in Florida than in Georgia.¹

Consequently, the general goal of most business intelligence systems was to: (1) access data from a variety of different sources; (2) transform these data into information, and then into knowledge; and (3) provide an easy-to-use graphical interface to display this knowledge. In other words, a business intelligence system was responsible for collecting and digesting data, and presenting knowledge in a friendly way (thus enhancing the end-user’s ability to make good decisions). The following diagram illustrates the processes that underpin a traditional business intelligence system:

¹ Note that *business intelligence* can be defined both as a “state” (a report that contains knowledge) and a “process” (software responsible for converting data into knowledge).



Although different texts have illustrated the relationship between data and knowledge in different ways, the distinction between data, information, and knowledge is quite clear:

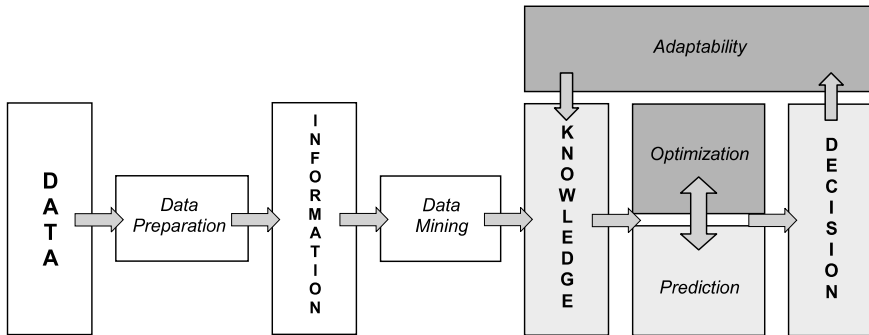
- *Data* are collected on a daily basis in the form of bits, numbers, symbols, and “objects.”
- *Information* is “organized data,” which are preprocessed, cleaned, arranged into structures, and stripped of redundancy.
- *Knowledge* is “integrated information,” which includes facts and relationships that have been perceived, discovered, or learned.

Because knowledge is such an essential component of any decision-making process (as the old saying goes, “*Knowledge is power!*”), many businesses have viewed knowledge as the final objective. But it seems that knowledge is no longer enough. A business may “know” a lot about its customers – it may have hundreds of charts and graphs that organize its customers by age, preferences, geographical location, and sales history – but management may still be unsure of what decision to make! And here lies the difference between “decision support” and “decision making”: all the knowledge in the world will not guarantee the right or best decision.

Moreover, recent research in psychology indicates that widely held beliefs can actually hamper the decision-making process. For example, common beliefs like “the more knowledge we have, the better our decisions will be,” or “we can distinguish between useful and irrelevant knowledge,” are not supported by empirical evidence. Having more knowledge merely increases our confidence, but it does not improve the accuracy of our decisions. Similarly, people supplied with “good” and “bad” knowledge often have trouble distinguishing between the two, proving that irrelevant knowledge decreases our decision-making effectiveness.

Today, most business managers realize that a gap exists between having the right knowledge and making the right decision. Because this gap affects management’s ability to answer fundamental business questions (such as “What should be done to increase profits? Reduce costs? Or increase market share?”), the future of business intelligence lies in systems that can provide answers and recommendations, rather than mounds of knowledge in the form of reports. *The future of business intelligence lies in systems that can make decisions!* As a result, there is a new trend emerging in the marketplace called *Adaptive Business Intelligence*.

In addition to performing the role of traditional business intelligence (transforming data into knowledge), Adaptive Business Intelligence also includes the decision-making process, which is based on prediction and optimization:



While *business intelligence* is often defined as “a broad category of application programs and technologies for gathering, storing, analyzing, and providing access to data,” the term *Adaptive Business Intelligence* can be defined as “the discipline of using prediction and optimization techniques to build self-learning ‘decisioning’ systems” (as the above diagram shows). Adaptive Business Intelligence systems include elements of data mining, predictive modeling, forecasting, optimization, and adaptability, and are used by business managers to make better decisions.

This relatively new approach to business intelligence is capable of recommending the best course of action (based on past data), but it does so in a very special way: An Adaptive Business Intelligence system incorporates prediction and optimization modules to recommend near-optimal decisions, and an “adaptability module” for improving future recommendations. Such systems can help business managers make decisions that increase efficiency, productivity, and competitiveness. Furthermore, the importance of *adaptability* cannot be overemphasized. After all, what is the point of using a software system that produces sub par schedules, inaccurate demand forecasts, and inferior logistic plans, time after time? Would it not be wonderful to use a software system that could *adapt* to changes in the marketplace? A software system that could *improve* with time?

The concept of adaptability is certainly gaining popularity, and not just in the software sector. Adaptability has already been introduced in everything from automatic car transmissions (which adapt their gear-change patterns to a driver’s driving style), to running shoes (which adapt their cushioning level to a runner’s size and stride), to Internet search engines (which adapt their search results to a user’s preferences and prior search history). These products are very appealing for individual consumers, because, despite their mass production, they are capable of adapting to the preferences of each unique owner after some period of time.

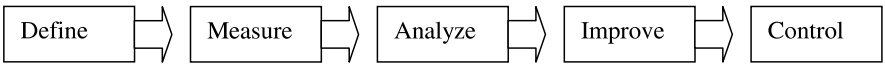
The growing popularity of adaptability is also underscored by a recent publication of the US Department of Defense. This lists 19 important research topics for the next decade and many of them include the term “adaptive”: *Adaptive Coordi-*

nated Control in the Multi-agent 3D Dynamic Battlefield, Control for *Adaptive* and Cooperative Systems, *Adaptive* System Interoperability, *Adaptive* Materials for Energy-Absorbing Structures, and Complex *Adaptive* Networks for Cooperative Control.

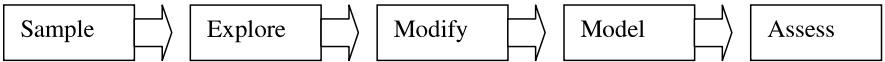
For sure, adaptability is here to stay. It is a vital component of any intelligent system, as it is hard to argue that a system is “intelligent” if it does not have the capacity to adapt. Moreover, modern definitions of natural and artificial intelligence include the term “adaptive.” For humans, the importance of adaptability is obvious: our ability to adapt was a key element in the evolutionary process. In the case of artificial intelligence, consider a chess program capable of beating the world chess master: Should we call this program intelligent? Probably not. We can attribute the program’s performance to its ability to evaluate the current board situation against a multitude of possible “future boards” before selecting the best move. However, because the program cannot learn or adapt to new rules, the program will lose its effectiveness if the rules of the game are changed or modified. Consequently, because the program is incapable of learning or adapting to new rules, the program is not intelligent.

The same holds true for any expert system. No one questions the usefulness of expert systems in some environments (which are usually well defined and static), but expert systems that are incapable of learning and adapting should not be called “intelligent”! Some expert knowledge was programmed in, that is all.

It is not surprising that the fundamental components of Adaptive Business Intelligence are already emerging in other areas of business. For example, the *Six Sigma* methodology is a great example of a well-structured, data-driven methodology for eliminating defects, waste, and quality-control problems in many industries. This methodology recommends the following sequence of steps:



Note that the above sequence is very close “in spirit” to part of the previous diagram, as it describes (in more detail) the adaptability control loop. Clearly, we have to “measure,” “analyze,” and “improve,” as we operate in a dynamic environment, so the process of improvement is continuous. The SAS Institute proposes another methodology, which is more oriented towards data mining activities. Their methodology recommends the following sequence of steps:



Again, note that the above sequence is very close to another part of our diagram, as it describes (in more detail) the transformation from data to knowledge. It is not surprising that businesses are placing considerable emphasis on these areas, because better decisions usually translate into better financial performance. And

better financial performance is what Adaptive Business Intelligence is all about. Systems based on Adaptive Business Intelligence aim at solving real-world business problems that have complex constraints, are set in time-changing environments, have several (possibly conflicting) objectives, and where the number of possible solutions is too large to enumerate. Solving these problems requires a system that incorporates modules for prediction, optimization, and adaptability. In the following chapters of this book, we will discuss these modules in detail, and see how they are combined to create an Adaptive Business Intelligence system.

2 Characteristics of Complex Business Problems

“See the foxhound with hanging ears and drooping tail as it lolls about the kennels, and compare it with the same hound as, with gleaming eyes and straining muscles, it runs upon a breast-high scent – such was the change in Holmes since the morning.”

The Adventure of the Bruce-Partington Plans

“‘I had,’ said he, ‘come to an entirely erroneous conclusion which shows, my dear Watson, how dangerous it always is to reason from insufficient data.’”

The Adventure of the Speckled Band

The statement “complex business problems are difficult to solve” is so obvious that it does not require any justification. A closer look at any real-world business problem, whether in distribution, customer retention, or fraud detection, will bear witness to this obvious truth. Most complex business problems share the following characteristics, which is the reason they are so difficult to solve:

- *The number of possible solutions is so large that it precludes a complete search for the best answer.* In other words, the number of possible distributions, routes, fraud rules, or transportation plans is so large, that examining all the possibilities would take many centuries of supercomputing time.
- *The problem exists in a time-changing environment.* This means that yesterday’s decision, however optimal, may be far from optimal today.
- *The problem is heavily constrained.* For most problems, the final solution should satisfy many restrictions imposed by internal regulations, capacities, laws, and/or preferences. Sometimes finding even one feasible solution (i. e., a solution that satisfies all problem-specific constraints) is quite difficult.
- *There are many (possibly conflicting) objectives.* For example, the goal of many scheduling problems is to minimize both time and cost, but these two objectives work against each other (as a decrease in time usually results in an increase in cost, and vice versa). To allow business managers to effectively control these tradeoffs, such problems may require an entire set of solutions (rather than just a single solution).

Of course, the above list can be extended to include many other characteristics, such as incomplete information (e. g., the necessary data were not recorded), noisy data (e. g., the data contain rounded figures and estimates), and uncertainly (e. g.,

the data are not reliable). However, the four primary characteristics listed above are sufficient for our purposes, so let us discuss each of them in turn.

2.1 Number of Possible Solutions

Let us assume we want to find the best solution to a problem with 100 decision variables. To keep this example simple, let us also assume that each of these decision variables is binary (i. e., each decision variable takes one of two possible values, such as “yes” or “no”). Each possible combination of these 100 variables produces some result, which we can evaluate and label with a “quality measure score.” Assume, for example, that a sequence

“yes” & “yes” & “no” & “no” & “no” & “yes” & “no” & ... & “yes”

produces a quality measure score of 17.3, whereas the sequence

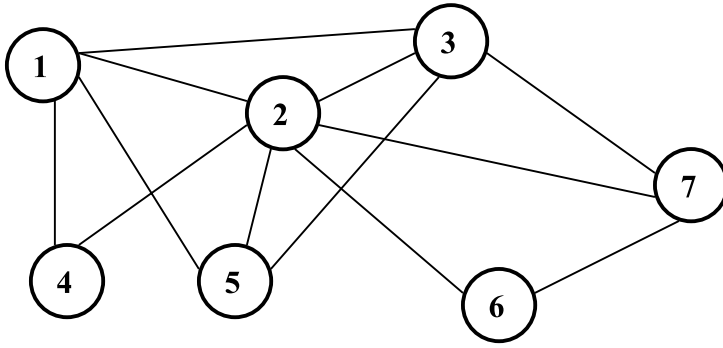
“yes” & “no” & “no” & “yes” & “no” & “yes” & “no” & ... & “no”

produces a quality measure score of 18.1. The higher the quality measure score, the better the solution, hence the latter solution is better than the former. Our task is to find the combination of values for the 100 variables that produces *the highest possible quality measure score*. In other words, we would like to find a solution that cannot be improved.

If we do not have any additional problem-specific knowledge, our approach might be to evaluate all possible combinations. However, the number of possible combinations is enormous. Although each variable can only take one of two values (“yes” or “no”), the number of possible solutions grows at an exponential rate: there are four combinations (2×2) for two variables, eight combinations ($2 \times 2 \times 2$) for three variables, and so on. With 100 variables, there are $2 \times 2 \times \dots \times 2$ (100 times) combinations – a number that corresponds to 10^{30} . Evaluating all of these combinations is simply impossible. Even if we had a computer that could test 1,000 combinations per second, and we began using this computer one billion years ago, we would have evaluated less than 1% of the possible solutions by today!

Let us consider another example, the famous traveling salesman problem. Conceptually, the problem is very simple: traveling the shortest possible distance, the salesman must visit every city in his territory (exactly once) and then return home.² The diagram below represents a seven-city version of this problem:

² Some closely related problems require slightly different criteria, such as finding a tour of the cities that yields the minimum travel time, minimum fuel cost, or a number of other possibilities, but the underlying principle is the same.



With seven cities, the problem has 360 possible solutions,³ making it relatively easy to solve. However, by adding a few more cities, this number also grows exponentially. To see the maddening growth of possible solutions, consider the following:

- A 10-city problem has 181,440 possible solutions.
- A 20-city problem has about 10^{16} possible solutions.
- A 50-city problem has about 10^{62} possible solutions.

By way of comparison, our planet holds approximately 10^{21} liters of water, so a 50-city problem has an unimaginably large number of solutions. The number of possible solutions to a 100-city problem exceeds (by many orders of magnitude) the estimated number of atoms in the whole Universe! It is so large that we cannot even conceive of sets with that many elements. Note also, that most real-world business problems are far more complex than this (in terms of possible solutions). They are defined by a much larger number of variables, and these variables usually take on more values than just “yes” or “no.” In such cases, the number of possible solutions is truly astronomical!

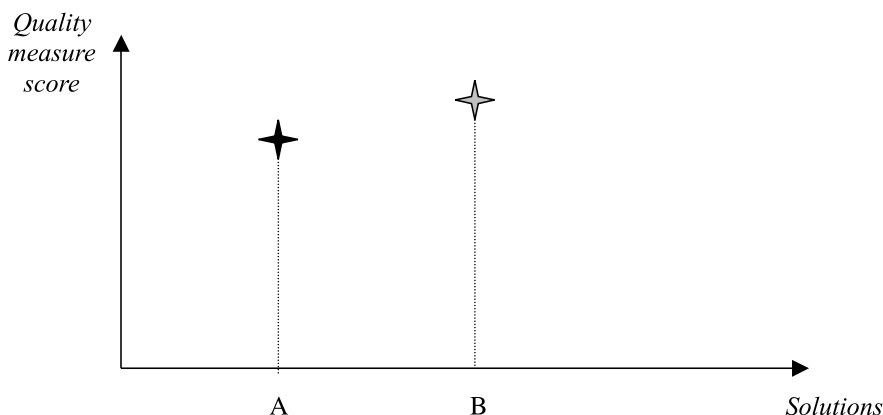
So, how can business managers find optimal solutions to such problems? Because the numbers of possible distributions, routes, fraud rules, or transportation plans might be so large that to examine all possibilities (with even the fastest supercomputers) would take many centuries at the best, an exhaustive search that relies on computing power is clearly not the answer. In the following chapters, we will discuss a real-world business problem where the number of possible solutions is *much* larger than the numbers presented here, and show how such problems can be solved using an Adaptive Business Intelligence system.

³ We assume that the problem is symmetric (i.e., the distance between cities i and j is the same as the distance between j and i). Note also, that solution 1–2–3–4–5–6–7 is the same as solution 3–4–5–6–7–1–2, as both these solutions have a different starting city but represent the same cycle.

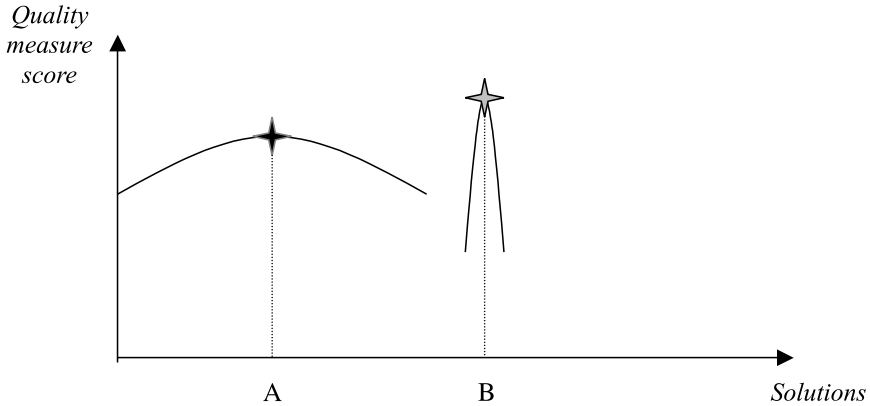
2.2 Time-Changing Environment

Business managers know that the marketplace is not static, and yet they take static snapshots of the problems they are trying to solve. A snapshot is a good starting point for analyzing and understanding a problem, but on its own, it paints a false picture. Because real-world business problems are set in time-changing environments, it is important to address the time factor explicitly. To illustrate this point, let us consider a real-world version of the traveling salesman problem with many delivery trucks. If the problem is carefully analyzed and a set of delivery routes found, the quality of these routes will be affected by many cyclical factors (such as rush-hour and weekend traffic, weather and road conditions, and so forth), and by random events (such as labor strikes or delivery truck accidents). Because the problem is influenced by so many environmental factors, any solution to a static snapshot of this problem might prove useless.

There are some additional issues related to time-changing environments that are worth noting. For example, imagine that we are considering the implementation of solution A or solution B:



Which of these two solutions would we select? Well, the question seems trivial: Because solution B has a higher quality measure score, solution B is better than solution A. Although this statement is true – solution B *is* better than solution A – the answer might not be that straightforward. It may happen that solution A “sits” on a relatively flat peak, whereas solution B “sits” on a very narrow peak:



We can interpret the above graph as follows: Solution B is better than solution A (there is no doubt about that), but if we are forced to modify solution B for any reason (because of equipment failure, bad weather, labor strike, etc.), then the quality of solution B will deteriorate very quickly. Solution A, on the other hand, is much more “stable” in the sense that it can tolerate changes and modifications without a sharp drop in quality. Given that solution A is less risky than solution B, should we still select the “better” solution B?

2.3 Problem-Specific Constraints

All real-world business problems have constraints of some sort, and if a particular solution does not satisfy these constraints then we cannot consider this solution. Scheduling problems serve as a good example of real-world constraints. For example, consider the problem of scheduling airline crews for various flights. First, we have to make a list of all the flights that require service, along with all the requirements for these flights (such as skill levels, number of crewmembers, and so forth). Next, we need a database of all available crewmembers, together with their preferences and characteristics. Once we have all this information, we need to find the “best” assignment of individual crewmembers to different flights – but what does the “best” mean? Well, in this case it may mean the cost of implementing the schedule, the flexibility of making changes/replacements, the degree to which personnel preferences are satisfied, etc. Note, however, that the final schedule must satisfy a few *hard constraints*, such as:

- A crewmember cannot serve on two flights at the same time.
- The crewmembers scheduled for a specific flight must satisfy some requirements (e. g., the captain must have clearance to operate that particular plane, the number of stewardesses should correspond to the capacity of the plane).
- Various laws and regulations, which might call for some minimum layover time for crewmembers in between flights.

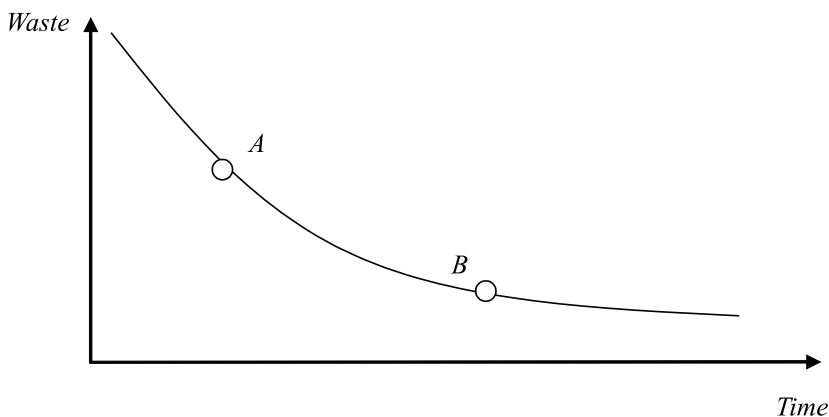
These are examples of hard constraints, which a feasible solution cannot violate. In addition to hard constraints, there are also many *soft constraints*. These represent requirements that are not mandatory, but “nice to have,” such as:

- Providing crewmembers with five consecutive rest days each month.
- Not scheduling some crewmembers together on the same flight (for personal reasons).
- Minimizing layover time.

Similar considerations apply to the traveling salesman problem: constraints include capacity limits, delivery time windows, maximum driving time, etc. Some of these constraints are hard (e. g., not transporting chemicals and food together on the same truck), while others are soft (e. g., personnel preferences). But regardless of whether we are routing trucks or scheduling crewmembers, it is necessary to assert the relative importance of each soft constraint by assigning numeric weights to it. When solving the problem, we can then use these weights to calculate a final quality measure score for each possible solution. Without first doing this, it would be extremely difficult to evaluate the various solutions.

2.4 Multi-objective Problems

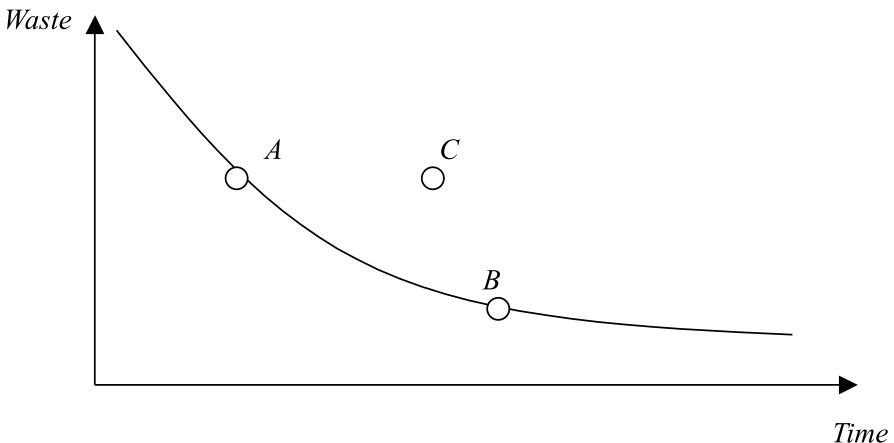
It is quite unusual for any real-world business problem to have only one objective. Consider, for example, what objectives are important for optimizing production? The objectives may include the minimization of production time and the minimization of material waste. Note, however, that these objectives might “work” against each other, as the minimization of production time may trigger an increase in material waste, and vice versa. We can illustrate this tradeoff with the following diagram:



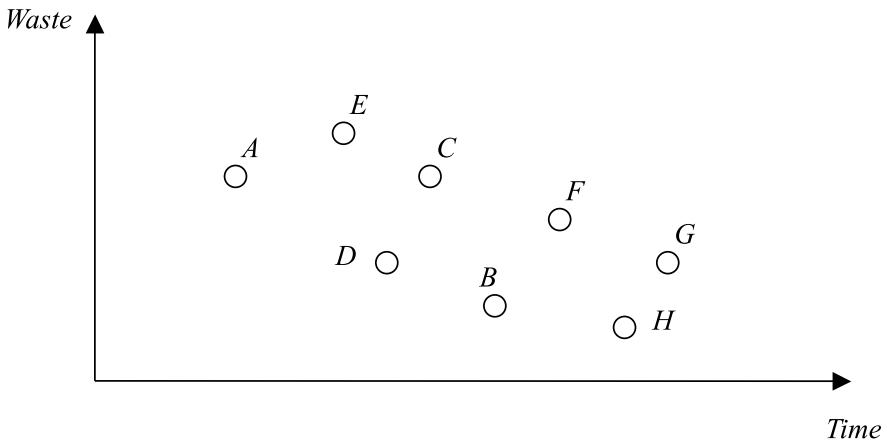
Clearly, the shorter the production time, the greater the waste. The above curve gives us the approximate relationship between these two objectives, and we can estimate the amount of additional time required to reduce waste by a specific amount. Let us consider solutions A and B: Which of them is better? Solution A is faster, but the amount of material waste is higher, and vice versa. In problems with multiple objectives, it is possible to find a solution that is best with respect to the first objective, but not the second, and a different solution that is best with respect to the second objective, but not the first. Multiple-objective (multi-objective) problems pose the challenge of defining the quality of a solution in terms of several (possibly conflicting) parameters.

So, which solution should we select? A or B? It is impossible to answer this question without first agreeing on a common denominator for time and waste. For example, we can translate both objectives into dollars by calculating that five minutes of production time is worth \$100, and each pound of material waste is worth \$180. We can then calculate the merits (expressed in dollars) of both solutions, compare the numbers, and select the solution with the lowest dollar figure. This approach can be used with a larger number of objectives, but it might be tricky to find the common denominator for some of these objectives if they include criteria such as “workplace health and safety.”

The real-world version of the traveling salesman problem also has multiple objectives: Besides trying to minimize the total travel distance, we are also trying to make as many “on time” deliveries as possible, balance the travel time among all the delivery trucks, and so on. If we have a set of possible solutions to this multiple-objective problem, then it might be convenient to classify these solutions into *dominated* solutions and *non-dominated* solutions. A solution is dominated if a feasible solution exists (i. e., a solution that satisfies all problem-specific constraints) that is (1) at least as good with respect to every objective, and (2) strictly better with respect to at least one objective. The figure below illustrates the case: Solution C is dominated because solution A is as good as solution C on the *waste* dimension, and better on the *time* dimension:



A solution that is not dominated by any other feasible solution is called a *non-dominated* solution. Again, the following figure illustrates the case, where solutions E, C, F, and G are dominated:



Solution E is dominated because A is better on all objectives; solution C is dominated because D is better on all objectives; and solutions F and G are dominated because B is better on all objectives. On the other hand, solutions A, D, B, and H are non-dominated, because no solutions exist that are as good or better than any of them on all objectives.

Of course, the non-dominated solutions are of interest to us. Ideally, any system that deals with multiple-objective problems should return several diverse solutions (preferably all non-dominated). Each of these solutions might be of interest to us, but in most cases, we can only implement one solution. To decide between these various non-dominated solutions, either human expertise is used to find common denominators (e. g., to express time and waste in dollars) or some higher-level knowledge.

For example, we may assess the relative importance of each objective by assigning numeric weights (in the same manner they were applied to soft constraints in the previous section), or by imposing a ranking for all objectives and then selecting solutions that follow this ranking. Another option would be to select the most important objective and then convert the remaining objectives into constraints that must be satisfied. Very often, the selected approach is dependent on the problem.

2.5 Modeling the Problem

The problem-solving process consists of two separate steps: (1) creating a model of the problem, and (2) using that model to generate a solution: